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Resumen/Summary The purpose of this Technical Report is to define the way to use the INTEGRAL Program. This program analyses the case of a crack in an integrally stiffened panel in a direction that is perpendicular to it. Also considered in the report are the stiffeners situated parallel to the direction of the propagation, which serve as a rigid support to them. This assumed configuration corresponds to the practical case of an integrally stiffened panel. This set is, in turn, supported on many more rigid elements, such as ribs or frames. Even though this program is only applicable to symmetric configurations, it can also be used in configurations where the asymmetric arrangements are not significant. PROPIEDAD DE EADS CASA / AIRBUS ESPAÑA. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Proyectos y Sistemas de EADS CASA / AIRBUS ESPAÑA. EADS CASA / AIRBUS ESPAÑA PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorisation by EADS CASA / AIRBUS ESPAÑA Engineering and System Direction.				
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REGISTRO DE REVISIONES/REVISIONS RECORD

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1.- Technical sheet of the program

Technical Report **NT-T-ADP-01044**

<u>PROGRAM</u>	<u>REFERENCE</u>	<u>DATE</u>
INTEGRAL		2003

OBJECTIVE

This program analyses the case of a crack in an integrally axial stiffened panel in a direction that is perpendicular to it.

LANGUAGE

Fortran 77

COMPUTER

Sun Solaris

NUMBER OF LINES (APROX.)

3400

EXECUTION MODE

Interactive

AUTHOR

Darío Fernández Villalba

VERSION

2.2

ADDITIONAL DOCUMENTS

Technical Report: NT-T-ADP-02016. Issue A
Theoretical Manual of the INTEGRAL Program.

Technical Report: NT-T-ADP-02007. Issue A
Programmer's Manual of the INTEGRAL Program.

2.- Introduction

This note makes up the User's Manual for the INTEGRAL Program, which has been developed by EADS CASA from 1993 to date.

This program analyses the case of a crack in an integrally axial stiffened panel in a direction that is perpendicular to it. Also considered in the report are the stiffeners parallel to the direction of the propagation, which serve as rigid support for them.

The program can be executed in the "Technical Calculation Computer" (Sun Solaris), as described in section 3.

The data required for its execution, as well as the format for the required input file, are specified in section 4.

In addition, the results, as well as the possible errors in execution, are included in the corresponding output file, as shown in section 5.

In parallel with this program, a Windows application has been developed using visual environment as described in section 6.

Finally, in chapter 7, two examples are included which can clarify the procedures for execution and the different options offered by the program.

Differences with previous version 2.0:

Now, the user can choose the language for screen dialogs and for output files, between Spanish and English. In previous versions, the screen dialogs were in Spanish, meanwhile the output files were written in English.

3.- Execution of the program

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
$integral
```

The program asks for the language to use and the name of the input and output files, as shown in the following example:

```
Idioma / Language:
=====

  1 ... ESPAÑOL

  2 ... ENGLISH

Idioma / Language: 2

Name of the input file: Example.dat

Name of the output file: Example.res
```

The CPU time for calculation depends on the number of stiffeners, being a few seconds as much.

It is possible to execute the program with a single command, including the input and output filenames and the language as parameters. The above example can be executed as follows:

```
$integral Example.dat Example.res 2
```

4.- Input data file

This file contains all the necessary data that define the problem, as well as the different options required for solving it.

Basically, it contains eight data groups, all of which are necessary. They will be numbered from 1 to 8 and should appear in strict order, from lower to higher.

Each set of 8 groups makes up the solution to a separate problem. If we want the solution for different problems, they must appear in the input file, as many sets as well as number of different problems.

These sets will be included one after the other. Each set of problem data will contain only the data groups that are different from the previous one.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numeration of each group will be included in the first column.

Listed below are the contents of each data group:

GROUP 1: Title of the problem.

Four lines making up the title will be included (66 characters at most), which will appear at the head of each page of the output file:

Line 1 -----> 1 First line of the title

Line 2 -----> Second line of the title

Line 3 -----> Third line of the title

Line 4 -----> Fourth line of the title

GROUP 2: Data on the skin.

The structure of the data of this group is the following:

Line 1 -----> 2 Optional commentary"

Line 2 -----> E μ t Kc L

Where: E = Young modulus of the skin.

μ = Poisson modulus

t = Panel thickness

Kc = Fracture toughness.

L = Distance between lateral stiffeners. If there is none (infinite spacing), it will be given as data L = 0.

GROUP 3: Data on the stiffeners.

The structure of the data of this group is the following:

Line 1 ----->	3	Optional commentary			
Line 2 ----->	nl	X(l)	X(2)	...	X(nl)
Line 3 ----->		ts(l)	ts(2)	...	ts(nl)
Line 4 ----->		h (l)	h (2)	...	h (nl)
Line 5 ----->		A (l)	A (2)	...	A (nl)
Line 6 ----->		I (l)	I (2)	...	I (nl)
Line 7 ----->		Zo(l)	Zo(2)	...	Zo(nl)
Line 8 ----->		Zi(l)	Zi(2)	...	Zi(nl)

Where: nl = Number of axial stiffeners.

X(i) = Co-ordinate X of each stiffener.

ts(i) = Thickness of the stiffener in the section of the attachment to the skin.

h(i) = Height of the stiffener or distance of the propagation of the crack, from which it reaches the stiffener, until it breaks entirely.

A(i) = Area of the stiffener.

I(i) = Moment of inertia.

Zo(i) = Distance from the centre of gravity to the exterior line (joined to the skin)

Zi(i) = Distance from the center of gravity to the interior line (free line).

The stiffeners should be given in such a way that their coordinate X is arranged from less to more.

GROUP 4: Data on the joint to the skin of each stiffener.

As has been showed in Ref. 1, the stiffeners, due to the fact that they are integral, are joined to the skin in a continuous manner, but, for the analysis, it will be assumed that they are joined by means of a finite number of attachment points.

Two types of attachment system can be defined between the skin and the stiffeners, uniform or variable, as desired.

A UNIFORM attachment is the one in which all the attachment points of a stiffener are defined, aligned all along one or several lines, defined in accordance with the coordinate X of the stiffener, and the distance between the lines and the attachment points are uniformly spaced.

In this case of UNIFORM attachment system, the structure of this Group will be the following:

```

Line 1      ----->  4  UNI Optional commentary

[nl] lines  ----->   nx(1)   ny(1)   pitch(1)   Wm(1)

                        nx(2)   ny(2)   pitch(2)   Wm(2)

                        .....   .....   .....   .....

                        .....   .....   .....   .....

                        nx(nl)  ny(nl)  pitch(nl)  Wm(nl)

```

A VARIABLE attachment system is the one in which the attachment points of one stiffener (or of all the stiffeners) does not comply with any of the conditions imposed for the Uniform attachment system. In this case, the structure of the group would be the following:

```

Line 1      ----->  4  VAR Optional commentary

[nu(1)] lines ----->   nu(1)   Xu(1,1)   Yu(1,1)   W(1,1)

                        Xu(1,2)   Yu(1,2)   W(1,2)

                        .....   .....   .....

                        .....   .....   .....

                        Xu(1,nr)  Yu(1,nr)  W(1,nr)

[nu(2)] lines ----->   nu(2)   Xu(2,1)   Yu(2,1)   W(2,1)

                        Xu(2,2)   Yu(2,2)   W(2,2)

                        .....   .....   .....

                        .....   .....   .....

                        Xu(2,nu)  Yu(2,nu)  W(2,nu)

.....

.....

```

And, successively, with as many sets of lines, as there are stiffeners.

The significance of each one of the prior magnitudes, defined for the case of a uniform joint, is the following:

- nx(i) = Number of lines of attachment points defined for the stiffener “i”.
- ny(i) = Number of attachment points defined in each line of junction of the stiffener “i”.
- Pitch(i) = Spacing of the attachment points along each line.

$W_m(i)$ = Portion of the thickness of the stiffener on which the stress of each attachment point is distributed. If it is defined as zero, the program calculates it in an automatic manner as follows:

$$W_m = 0 \implies W_m = t_s/n_x$$

In the case of a variable attachment system, the significance of each magnitude is:

$nu(i)$ = Number of attachment points for the stiffener “i”.

$X_u(i, j)$ = Coordinate X of the point “j” of the stiffener “i”.

$Y_u(i, j)$ = Coordinate Y of the point.

$W(i, j)$ = Portion of the thickness of the stiffener on which the stress of the attachment point is distributed.

GROUP 5: Stresses in the far field.

The structure of the data of this group is the following:

Line 1 -----> 5 Optional commentary

Line 2 -----> σ R

Where: σ = Axial stress of reference in the far field. This stress is defined as the axial load applied to the set of panels and stiffeners, divided by their total area.

R = Ratio between the transversal stress σ_x and the reference stress σ .

GROUP 6: Lengths of the crack.

The structure of the data of this group is the following:

Line 1 -----> 6 Optional commentary

Line 2 -----> na a(1) a(2) ... a(na)

Where: na = Number of lengths of the crack to be analysed.

a(i) = Each one of the “na” lengths of the crack.

It is not necessary for the lengths of the crack to be defined in an orderly manner, for the program will put them in order automatically from smaller to larger.

In addition to the lengths of the crack for the analysis which have been defined by the user, the program will automatically include (if they have not been defined already), those for which the crack tip reaches a stiffener (a1), or it ends up breaking entirely (a2).

These lengths, for each stiffener, are the following:

$$a_1(i) = X(i) - \frac{ts(i)}{2}$$

$$a_2(i) = X(i) + h(i)$$

GROUP 7: Print control.

The structure of the data of this group is the following:

Line 1 -----> 7 Optional commentary

Line 2 -----> I(1) I(2) I(3)

Where: I(1) = Printing code for the results of the analysis for the skin.

I(2) = Printing code for the results of the analysis for the stiffeners.

I(3) = Printing code for the results of the analysis for the stress on the attachment points.

These codes are defined as follows:

I(i) = 0: The corresponding results are not written in the output file.

I(i) = 1: The corresponding results are written in the output file.

GROUP 8: Execution of the program.

This group invokes the execution of the program and should appear as many times as problems there are to resolve.

The structure of the data in this group is the following:

Line 1 -----> 8 Optional commentary

5.- Output results

The INTEGRAL program creates a single output file. This file contains the results of the analysis for the different configurations and crack lengths, which have been specified in the input file.

This file presents the following information for each one of the analyses of the program:

- Summary of the input data:
 - = Data of the skin.
 - = Data of the stiffeners.
 - = Data of the joint of the stiffeners to the skin.
 - = Data of the far field stress.
- Results for the skin. For each crack length:
 - = Stress distribution in the far field (for $X = a$)
 - = Stress intensity factor.
 - = Factor β (normalised stress intensity factor).
 - = Residual strength.
 - = State of each stiffener (Intact or broken).
- Results for the stiffeners. For each crack length:
 - = Stress in the far field (regardless of the length of the crack).
 - = Axial load.
 - = Bending moment.
 - = Stress on the extreme lines of the stiffener.
 - = Axial stress of the stiffener.
- Load transferred by the attachment points between skin and stiffeners, for each crack length.
- Errors detected in the input data.

In Chapter 7, two examples are given of the output file.

When the program detects an error in the input data, it writes it on the results file and interrupts the running of the program. The error messages, which might appear, are detailed below:

Error messages in the input data. English output:

ERROR: The input file cannot be opened

ERROR: The output file cannot be opened

ERROR: Bad input data in input file

ERROR: Too many stiffeners exceeding the program capability

ERROR: Too many crack lengths for analysis exceeding the program capability

ERROR: Too many attachment points for a stiffener exceeding the program capability

ERROR: Group 1 missing in input file (Title)

ERROR: End of file while reading group 1 in input file (Title)

ERROR: Group 2 missing in input file (Skin)

ERROR: End of file while reading group 2 in input file (Skin)

ERROR: Group 2 is defined badly (Skin)

ERROR: Group 3 missing in input file (Stringers)

ERROR: End of file while reading group 3 in input file (Stringers)

ERROR: Group 3 is defined badly (Stringers)

ERROR: Group 4 missing in input file (Attachment points)

ERROR: End of file while reading group 4 in input file (Attachment points)

ERROR: Group 4 is defined badly (Attachment points)

ERROR: Group 4 (Attachment points) can not be defined before group 3 (Stringers)

ERROR: Unknown type of attachment points distribution

ERROR: Group 5 missing in input file (Stress field)

ERROR: End of file while reading group 5 in input file (Stress field)

ERROR: Group 5 is defined badly (Stress field)

ERROR: Group 6 missing in input file (Crack lengths)

ERROR: End of file while reading group 6 in input file (Crack lengths)

ERROR: Group 6 is defined badly (Crack lengths)

ERROR: Group 7 missing in input file (Print control)

ERROR: End of file while reading group 7 in input file (Print control)

ERROR: Group 7 is defined badly (Print control)

Error messages in the input data. Spanish output:

ERROR: No se puede abrir el fichero de entrada

ERROR: No se puede abrir el fichero de salida

ERROR: Datos erróneos en el fichero de entrada

ERROR: Demasiados rigidizadores, excediendo la capacidad del programa

ERROR: Demasiadas longitudes de grieta para el análisis, excediendo la capacidad del programa

ERROR: Demasiados puntos de unión para un rigidizador, excediendo la capacidad del programa

ERROR: Grupo 1 ausente en el fichero de entrada (Título)

ERROR: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título)

ERROR: Grupo 2 ausente en el fichero de entrada (Panel)

ERROR: Fin de fichero mientras se leía el grupo 2 en el fichero de entrada (Panel)

ERROR: Grupo 2 mal definido (Revestimiento)

ERROR: Grupo 3 ausente en el fichero de entrada (Rigidizadores)

ERROR: Fin de fichero mientras se leía el grupo 3 en el fichero de entrada (Rigidizadores)

ERROR: Grupo 3 mal definido (Rigidizadores)

ERROR: Grupo 4 ausente en el fichero de entrada (Puntos de unión)

ERROR: Fin de fichero mientras se leía el grupo 4 en el fichero de entrada (Puntos de unión)

ERROR: Grupo 4 mal definido (Puntos de unión)

ERROR: El grupo 4 (Puntos de unión) no puede ser definido delante del grupo 3 (Rigidizadores)

ERROR: Tipo desconocido de distribución de puntos de unión

ERROR: Grupo 5 ausente en el fichero de entrada (Campo de esfuerzos)

ERROR: Fin de fichero mientras se leía el grupo 5 en el fichero de entrada (Campo de esfuerzos)

ERROR: Grupo 5 mal definido (Campo de esfuerzos)

ERROR: Grupo 6 ausente en el fichero de entrada (Longitudes de grieta)

ERROR: Fin de fichero mientras se leía el grupo 6 en el fichero de entrada (Longitudes de grieta)

ERROR: Grupo 6 mal definido (Longitudes de grieta)

ERROR: Grupo 7 ausente en el fichero de entrada (Control de impresión)

ERROR: Fin de fichero mientras se leía el grupo 7 en el fichero de entrada (Control de impresión)

ERROR: Grupo 7 mal definido (Control de impresión)

6.- Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager.

The application can be executed by "double click" in the file **Integral.exe**, located in the PC's server, or by means a **link** on the "desktop" or in the "start menu".

The input data must be typed in the application form or retrieved from an input file. The form, after loading the data corresponding to the example 1 (see section 7), has the following appearance:

Integral - C:\Temp\Example1.dat

Run 1 | Run 2

☒ Title: Program INTEGRAL. Example 1.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 1 row with 20 attach points per stringer.

☒ Panel data: E: 73800.00 Mu: 0.33 t: 2.00 Kc: 3800.00 L: 500.00

☒ Stringers data:

	X	t	h	A	Ixx	Zo	Zi	No. of stringers
1	50.00	3.00	50.00	150.00	32250.00	25.00	25.00	2
2	150.00	3.00	50.00	150.00	32250.00	25.00	25.00	

☒ Joining points data: Joining type: Uniform

	No.(x)	No.(y)	Paso	W
Stringer 1	1	20	3.00	3.00

☒ Crack lengths:

	No. of crack lengths
1 0.00 4 30.00 7 60.00 10 90.00	21
2 10.00 5 40.00 8 70.00 11 100.00	
3 20.00 6 50.00 9 80.00 12 110.00	

Name of the output file: C:\Temp\Example1.lis

Name of the input file:

Language: Screen windows: ☐ Español ☒ English Output files: ☐ Español ☒ English

«New run» «Remove run» Program execution

Figure 1: Input data form.

Since this example includes two problems to be solved, the form has two tabs, one per problem, labelled "Run 1" and "Run 2", as shown above.

The form shows the data corresponding to the selected tab ("Run 1" in the above figure). The data corresponding to the second problem can be seen clicking on the tab "Run 2".

The program can be executed clicking the button “Program execution”. Then the application starts to solve the problems showing the following form:

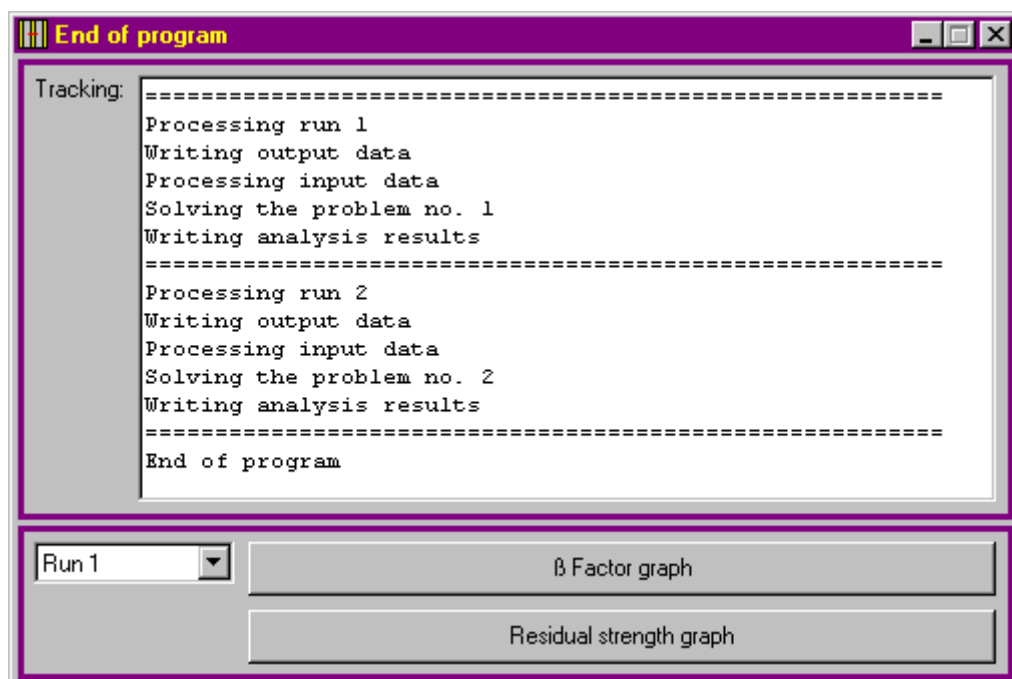


Figure 2: Tracking and results form.

The output results are written in the output file, but it is possible to see some of them in graphic way. For this purpose, there are two buttons to show the graphic of beta factor or residual strength. These buttons are only visible when the program execution has finished.

The graphics corresponding to the first problem of the above example (Example 1, detailed in section 7) are shown below:

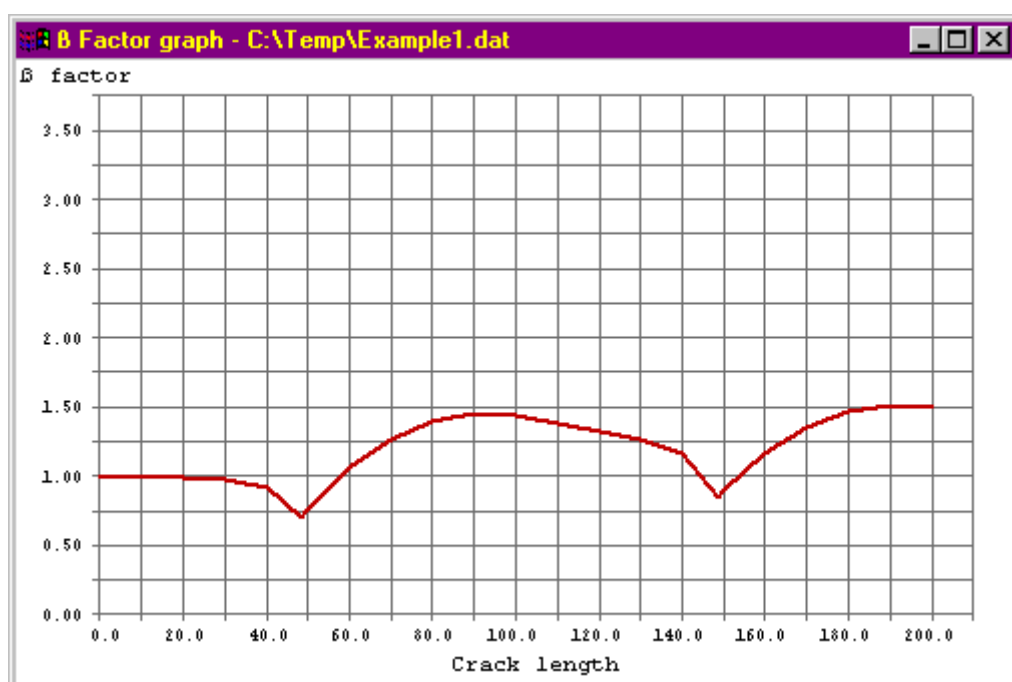


Figure 3: Beta factor graph.

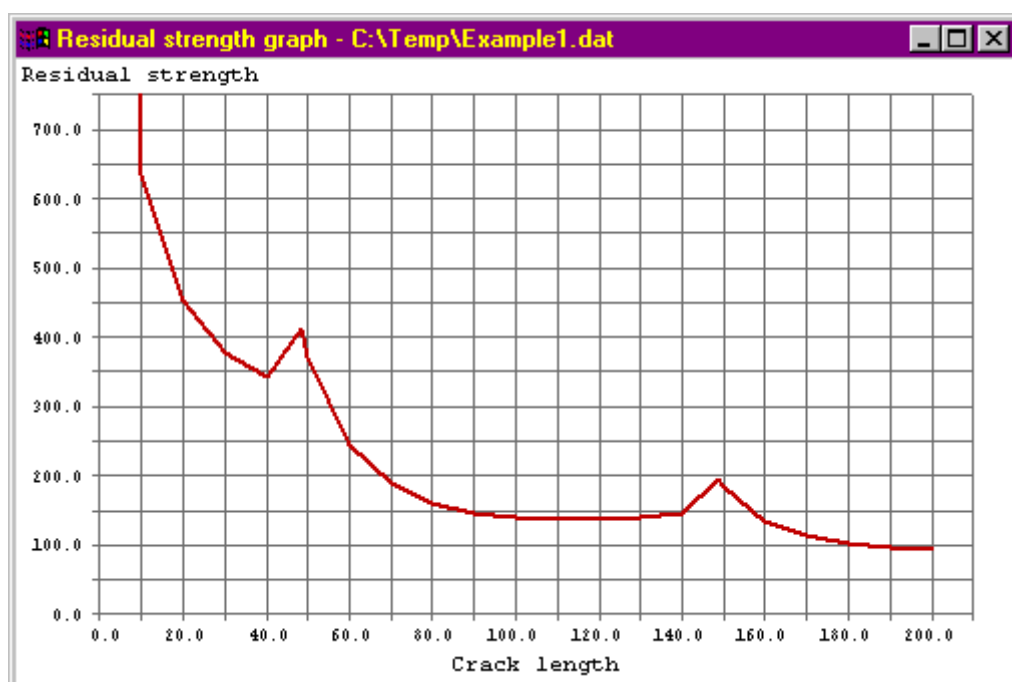


Figure 4: Residual strength graph.

7.- Examples

Two examples are given below for the use of the program; both of them correspond to the same problem, presented in different forms.

Formulation of the example problem:

The problem consists of a panel of material 2024 T351, with a basic thickness, equal to 2 mm, with a crack in the origin of the coordinates.

The panel is integrally stiffened by means of machined stringers of constant section, one of them is located at $X = 50$ mm, and the rest are spaced at every 100 mm in a uniform manner. The section common to all the stringers, together with the skin, is the following one:

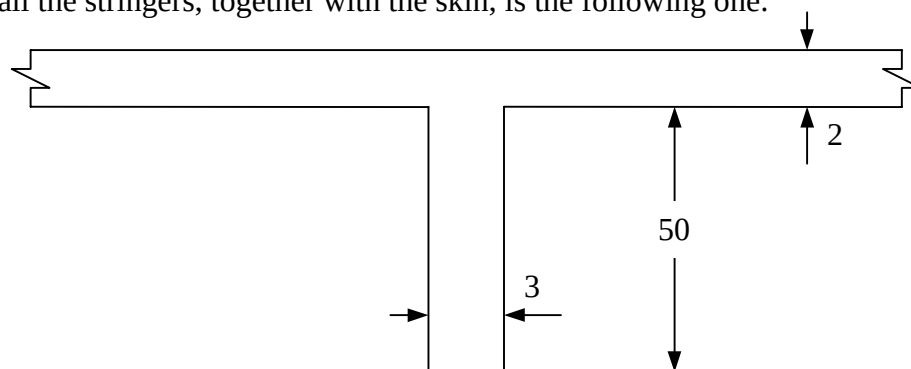


Figure 5: Example 1. Stringer geometry.

The stringers, in turn, are resting on frames, which are supposed to be infinitely rigid. One of them coincides with the direction of propagation of the crack and they are all spaced at every 500 mm.

The set of panel – stringers is submitted to a reference limit of axial stress in the far field, of 200 N/mm^2 . The stress in the normal direction to the stringers is half of the reference one.

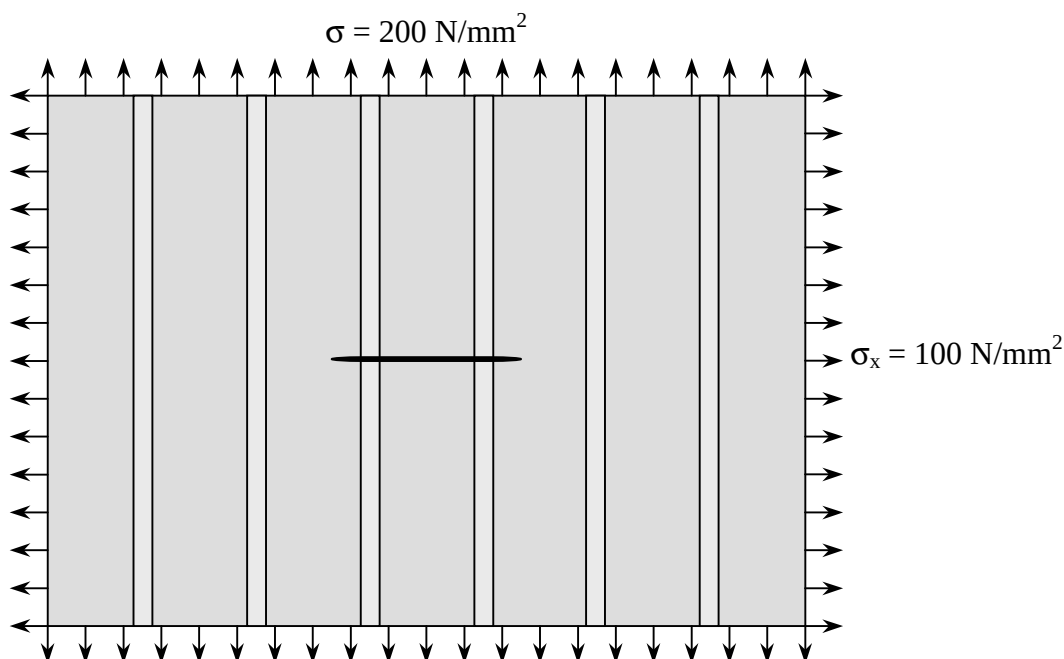


Figure 6: Example 1. General arrangement.

The data on the skin that must be introduced in the program, are the following:

- Young modulus $E = 73800 \text{ N/mm}^2$
- Poisson modulus..... $\mu = 0.33$
- Thickness..... $t = 2 \text{ mm}$
- Fracture toughness..... $K_c = 3800 \text{ N/mm}^2 \cdot \sqrt{\text{mm}}$
- Spacing of lateral stiffeners..... $L = 500 \text{ mm}$

The stringers are all of them equal, with the following properties:

- Thickness in the section of the joint..... $t_s = 3 \text{ mm}$
- Height $h = 50 \text{ mm}$
- Area $A = 150 \text{ mm}^2$
- Moment of inertia..... $I = 32250 \text{ mm}^4$
- Distance of CG to the exterior edge $Z_o = 25 \text{ mm}$
- Distance of the CG to the interior edge... $Z_i = 25 \text{ mm}$

The far field of stress will be defined by the following:

- Reference stress..... $\sigma = 200 \text{ N/mm}^2$
- Axial – lateral ratio..... $R = 0.5$

The analysis will be carried out for half crack lengths between 0 and 200 mm. Therefore, it will only be necessary to consider the effect of the stringers located at: $X = 50$, $X = 100$ and $X = 150 \text{ mm}$. Due to the symmetry, the ones located at $X = -50$ and $X = -150 \text{ mm}$ will implicitly be considered:

- Number de stiffeners $n_l = 2$
- Co-ordinates of the stiffeners $X = 50 - 150 \text{ mm}$
- Number of points of analysis..... $n_a = 21$
- Half crack lengths..... $a =$

0	10	20	30	40	50
	60	70	80	90	100
	110	120	130	140	150
	160	170	180	190	200

Example 1

In this example, the previous problem will be solved with the following additional considerations, in the two steps of the program:

– First step:

Twenty attachment points (at each side of the crack) will be considered for the transfer of the load between a stiffener and the skin. All of them will be aligned along the stiffener and spaced uniformly at every 3 mm. and so the data input will be specified in a uniform joint.

The load of each attachment point will be distributed all along the thickness of the stiffener (3 mm.).

The results requested will be all the possible ones, that is, the results for the skin, for the stiffeners and for all the attachment points.

– Second step:

Ten attachment points (at each side of the crack) will be considered for the transfer of the load between a stiffener and the skin.

A variable attachment system will be specified for the data input, but with the same distribution as for the first step.

Results will only be requested for the skin.

The difference between both steps lies in the definition of the junction between stiffeners and skin, although it will formally be the same in both cases, except for the number of points to be considered.

Therefore, this example could serve to evaluate the influence of the total number of attachment points considered per stringer.

With this data, we can prepare the input file. This file as well as the output file obtained by running the program are shown on next pages:

Input file:

```

1 Program INTEGRAL. Example 1.
  Crack in an integrally stiffened panel.
  Material: 2024 T351.
  Uniform attachment. 1 row with 20 attach points per stringer.
2 SKIN DATA: E  μ  t  Kc  L
73800.0      0.33      2.0      3800.0      500.0
3 STRINGERS DATA: nl  X  ts  h  A  I  Zo  Zi
2      50.0      150.0
      3.0      3.0
      50.0      50.0
      150.0      150.0
      32250.0      32250.0
      25.0      25.0
      25.0      25.0
4 UNIFORM ATTACHMENT SYSTEM: nx  ny  Pitch  W
1 20 3.0 3.0
1 20 3.0 3.0
5 FAR FIELD STRESS: Reference_stress  Sx/Sy
200.00      0.50
6 CRACK LENGTHS
21 0.0 10.0 20.0 30.0 40.0 50.0
      60.0 70.0 80.0 90.0 100.0
      110.0 120.0 130.0 140.0 150.0
      160.0 170.0 180.0 190.0 200.0
7 PRINT CONTROL: Skin Stringers Forces
1 1 1
8 RUN THE PROGRAM
1 Program INTEGRAL. Example 1.
  Crack in an integrally stiffened panel.
  Material: 2024 T351.
  Variable attachment. 1 row with 10 attach points per stringer.
4 VARIABLE ATTACHMENT: nu  X(i)  Y(i)  W(i)
10      50.0      1.5      3.0
      50.0      4.5      3.0
      50.0      7.5      3.0
      50.0      10.5      3.0
      50.0      13.5      3.0
      50.0      16.5      3.0
      50.0      19.5      3.0
      50.0      22.5      3.0
      50.0      25.5      3.0
      50.0      28.5      3.0
10      150.0      1.5      3.0
      150.0      4.5      3.0
      150.0      7.5      3.0
      150.0      10.5      3.0
      150.0      13.5      3.0
      150.0      16.5      3.0
      150.0      19.5      3.0
      150.0      22.5      3.0
      150.0      25.5      3.0
      150.0      28.5      3.0
7 PRINT CONTROL: Skin Stringers Allowables Forces
1 0 0
8 RUN THE PROGRAM

```

Output file:

Program INTEGRAL. Example 1.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 1 row with 20 attach points per stringer.

PAGE 1

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

SKIN DATA:

```
=====
```

Young's modulus.....:	73800.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	3800.00
Lateral stiffener spacing.....:	500.00

APPLIED STRESSES:

```
=====
```

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

```
=====
```

	STIFFENER 1	STIFFENER 2
	-----	-----
X-coordinate.....:	50.00	150.00
Uniform attachment system:		
Number of rows.....:	1	1
Number of points per row..:	20	20
Attachment pitch.....:	3.00	3.00
Attachment width.....:	3.00	3.00
Thickness at attachment.....:	3.00	3.00
Mid line length.....:	50.00	50.00
Area.....:	150.00	150.00
Moment of inertia.....:	32250.00	32250.00
Distance from CG to outer cap:	25.00	25.00
Distance from CG to inner cap:	25.00	25.00

Output file: (cont.)

Program INTEGRAL. Example 1.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 1 row with 20 attach points per stringer.

PAGE 2

```
*****
*
*   <<< R E S U L T S   F O R   S K I N >>>
*
*****
```

APPLIED STRESSES:

```
-----
Average axial stress (YD).....:    200.00
Ratio of stresses (XD/YD).....:      0.50
Unit axial skin stress.....:    214.14
Unit lateral skin stress.....:    100.00
```

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	RESIDUAL STRENGTH	STRESS INTENSITY	STIFFENERS STATUS
-----	-----	-----	-----	-----	-----
0.00	214.14	1.00000	*****. **	0.00	II
10.00	214.14	0.99757	634.73	1197.36	II
20.00	214.14	0.98905	452.69	1678.85	II
30.00	214.14	0.96956	377.05	2015.65	II
40.00	214.14	0.92090	343.79	2210.65	II
48.50	214.14	0.69873	411.49	1846.96	II
50.00	214.14	0.75957	372.80	2038.61	TI
60.00	214.14	1.06221	243.36	3122.96	TI
70.00	214.14	1.27175	188.19	4038.57	TI
80.00	214.14	1.39561	160.41	4737.91	TI
90.00	214.14	1.44858	145.70	5216.04	TI
100.00	214.14	1.43277	139.75	5438.21	XI
110.00	214.14	1.37810	138.53	5485.99	XI
120.00	214.14	1.32280	138.18	5500.03	XI
130.00	214.14	1.25810	139.59	5444.58	XI
140.00	214.14	1.15908	146.00	5205.42	XI
148.50	214.14	0.84166	195.22	3892.97	XI
150.00	214.14	0.89654	182.36	4167.69	XT
160.00	214.14	1.16949	135.36	5614.83	XT
170.00	214.14	1.35847	113.05	6722.86	XT
180.00	214.14	1.47018	101.51	7486.63	XT
190.00	214.14	1.51796	95.70	7941.72	XT
200.00	214.14	1.50370	94.16	8071.53	XX

Beta factors refered to the unit axial skin stress.

Residual strength is based on the reference stress.

Key for stiffeners status: I=Intact
T=Transition
X=Failed

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 3

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X =  50.00 >>>
*
*****
```

Unit stiffener stress: 181.14

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	STRESSES		
			OUTER CAP	AXIAL	INNER CAP
0.00	27171.4	0.0	181.14	181.14	181.14
10.00	27312.7	2396.2	183.94	182.08	180.23
20.00	27779.4	10519.3	193.35	185.20	177.04
30.00	28741.1	28232.6	213.49	191.61	169.72
40.00	30750.6	69066.0	258.54	205.00	151.46
48.50	36160.7	194997.4	392.23	241.07	89.91
50.00	0.0	0.0	0.00	0.00	0.00
60.00	0.0	0.0	0.00	0.00	0.00
70.00	0.0	0.0	0.00	0.00	0.00
80.00	0.0	0.0	0.00	0.00	0.00
90.00	0.0	0.0	0.00	0.00	0.00
100.00	0.0	0.0	0.00	0.00	0.00
110.00	0.0	0.0	0.00	0.00	0.00
120.00	0.0	0.0	0.00	0.00	0.00
130.00	0.0	0.0	0.00	0.00	0.00
140.00	0.0	0.0	0.00	0.00	0.00
148.50	0.0	0.0	0.00	0.00	0.00
150.00	0.0	0.0	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 4

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X = 150.00 >>>
*
*****
```

Unit stiffener stress: 181.14

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	STRESSES		
			OUTER CAP	AXIAL	INNER CAP
0.00	27171.4	0.0	181.14	181.14	181.14
10.00	27183.3	197.3	181.37	181.22	181.07
20.00	27219.0	792.0	182.07	181.46	180.85
30.00	27279.0	1790.4	183.25	181.86	180.47
40.00	27362.6	3188.7	184.89	182.42	179.95
48.50	27442.5	4538.8	186.47	182.95	179.43
50.00	27730.2	9272.1	192.06	184.87	177.68
60.00	29033.9	30723.7	217.38	193.56	169.74
70.00	29855.8	44247.9	233.34	199.04	164.74
80.00	30395.1	53121.7	243.81	202.63	161.45
90.00	30745.5	58886.3	250.62	204.97	159.32
100.00	30879.0	61083.1	253.21	205.86	158.51
110.00	32090.4	81388.9	277.03	213.94	150.84
120.00	33713.6	109632.9	309.74	224.76	139.77
130.00	36345.7	158234.0	364.97	242.30	119.64
140.00	41324.9	259314.9	476.52	275.50	74.48
148.50	53728.8	549838.4	784.42	358.19	-68.04
150.00	0.0	0.0	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 5

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 0.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	0.0	0.0
2	0.0	0.0
3	0.0	0.0
4	0.0	0.0
5	0.0	0.0
6	0.0	0.0
7	0.0	0.0
8	0.0	0.0
9	0.0	0.0
10	0.0	0.0
11	0.0	0.0
12	0.0	0.0
13	0.0	0.0
14	0.0	0.0
15	0.0	0.0
16	0.0	0.0
17	0.0	0.0
18	0.0	0.0
19	0.0	0.0
20	0.0	0.0

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 6

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 10.000

A. POINT	X = 50.00	X = 150.00
1	-1.2	0.0
2	-3.3	0.0
3	-4.8	-0.1
4	-5.5	-0.1
5	-5.3	-0.1
6	-4.3	-0.1
7	-2.6	-0.1
8	-0.4	-0.1
9	1.9	-0.1
10	4.1	-0.1
11	6.2	-0.1
12	8.0	0.0
13	9.5	0.0
14	10.7	0.1
15	11.8	0.3
16	12.8	0.4
17	13.9	0.7
18	15.8	1.2
19	19.1	1.9
20	54.8	8.3

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 1 row with 20 attach points per stringer.

PAGE 7

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 20.000

A. POINT	X = 50.00	X = 150.00
1	-5.9	-0.1
2	-16.3	-0.2
3	-22.9	-0.3
4	-24.3	-0.4
5	-20.7	-0.5
6	-13.2	-0.5
7	-3.4	-0.6
8	7.2	-0.5
9	17.4	-0.5
10	26.4	-0.4
11	33.9	-0.3
12	39.7	-0.1
13	44.2	0.2
14	47.4	0.5
15	49.9	1.0
16	52.4	1.8
17	55.5	2.9
18	61.4	4.6
19	73.2	7.8
20	205.7	33.2

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 8

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 30.000

A. POINT	X = 50.00	X = 150.00
1	-21.2	-0.1
2	-54.5	-0.4
3	-65.6	-0.6
4	-53.4	-0.8
5	-25.9	-1.0
6	7.3	-1.2
7	39.1	-1.2
8	65.7	-1.2
9	86.0	-1.1
10	100.2	-1.0
11	109.2	-0.7
12	114.2	-0.2
13	116.3	0.4
14	116.7	1.3
15	116.4	2.4
16	116.6	4.1
17	119.2	6.5
18	127.9	10.5
19	148.6	17.5
20	406.9	74.4

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 9

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 40.000

A. POINT	X = 50.00	X = 150.00
1	-95.1	-0.2
2	-164.7	-0.6
3	-68.2	-1.1
4	67.5	-1.4
5	169.0	-1.7
6	228.0	-2.0
7	255.9	-2.1
8	264.3	-2.1
9	261.3	-1.9
10	252.0	-1.5
11	239.6	-1.0
12	226.0	-0.2
13	212.6	0.9
14	200.3	2.4
15	189.9	4.5
16	182.7	7.4
17	180.7	11.7
18	188.5	18.7
19	214.4	30.8
20	574.7	130.5

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 10

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 48.500

A. POINT	X = 50.00	X = 150.00
1	243.2	-0.3
2	1248.7	-0.8
3	1015.2	-1.4
4	823.9	-1.9
5	684.8	-2.3
6	580.6	-2.6
7	500.1	-2.7
8	436.1	-2.6
9	384.3	-2.3
10	341.5	-1.8
11	305.8	-1.0
12	275.8	0.1
13	250.8	1.7
14	230.1	3.8
15	213.9	6.6
16	202.5	10.7
17	197.9	16.6
18	204.6	26.4
19	231.4	43.3
20	618.2	181.6

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 11

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 50.000

A. POINT	X = 50.00	X = 150.00
1	175.2	-1.1
2	1094.0	-3.2
3	907.2	-5.2
4	731.9	-6.8
5	602.5	-7.9
6	504.3	-8.5
7	427.3	-8.4
8	364.9	-7.6
9	313.2	-6.3
10	269.2	-4.3
11	231.1	-1.6
12	197.4	1.7
13	167.0	5.8
14	138.6	11.0
15	111.3	17.5
16	83.4	26.1
17	52.7	38.4
18	14.1	58.1
19	-41.0	91.7
20	-382.0	369.4

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 12

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 60.000

A. POINT	X = 50.00	X = 150.00
1	-132.6	-4.7
2	393.2	-14.2
3	417.9	-22.7
4	315.1	-29.3
5	229.7	-33.6
6	158.9	-35.3
7	97.6	-34.2
8	42.2	-30.4
9	-9.1	-24.1
10	-58.4	-15.3
11	-107.3	-4.3
12	-157.9	9.0
13	-212.9	24.7
14	-276.1	43.5
15	-353.8	66.4
16	-456.6	96.0
17	-605.3	137.0
18	-849.6	201.7
19	-1275.7	311.4
20	-4915.4	1220.9

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 13

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 70.000

A. POINT	X = 50.00	X = 150.00
1	-326.7	-7.0
2	-48.6	-21.1
3	109.4	-33.6
4	52.3	-43.5
5	-5.4	-49.8
6	-58.9	-52.2
7	-110.3	-50.5
8	-161.3	-44.8
9	-212.3	-35.3
10	-265.0	-22.3
11	-320.6	-6.0
12	-382.0	13.6
13	-452.3	36.6
14	-537.6	64.0
15	-647.0	97.3
16	-797.1	140.0
17	-1020.2	199.1
18	-1394.1	292.2
19	-2054.1	449.9
20	-7773.4	1757.7

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 14

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 80.000

A. POINT	X = 50.00	X = 150.00
1	-454.0	-8.5
2	-338.5	-25.6
3	-93.0	-40.9
4	-120.1	-52.8
5	-159.6	-60.5
6	-201.8	-63.3
7	-246.7	-61.2
8	-294.8	-54.2
9	-345.6	-42.6
10	-400.5	-26.8
11	-460.6	-7.1
12	-529.0	16.6
13	-609.4	44.5
14	-709.2	77.4
15	-839.4	117.6
16	-1020.4	168.9
17	-1292.4	239.9
18	-1751.4	351.6
19	-2564.8	540.8
20	-9648.7	2109.9

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 15

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 90.000

A. POINT	X = 50.00	X = 150.00
1	-536.7	-9.5
2	-526.8	-28.5
3	-224.5	-45.5
4	-232.1	-58.8
5	-259.8	-67.4
6	-294.6	-70.5
7	-335.3	-68.2
8	-381.6	-60.3
9	-432.2	-47.4
10	-488.6	-29.8
11	-551.6	-7.8
12	-624.4	18.6
13	-711.5	49.5
14	-820.7	86.2
15	-964.3	130.7
16	-1165.5	187.7
17	-1469.2	266.4
18	-1983.5	390.2
19	-2896.6	599.8
20	-10866.9	2338.7

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 16

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 100.000

A. POINT	X = 50.00	X = 150.00
1	-568.3	-9.8
2	-598.6	-29.7
3	-274.6	-47.3
4	-274.8	-61.1
5	-298.0	-70.0
6	-330.0	-73.3
7	-369.1	-70.8
8	-414.6	-62.7
9	-465.3	-49.2
10	-522.1	-30.9
11	-586.2	-8.0
12	-660.8	19.3
13	-750.4	51.5
14	-863.1	89.5
15	-1012.0	135.7
16	-1220.8	194.8
17	-1536.6	276.5
18	-2072.0	404.9
19	-3023.0	622.3
20	-11331.2	2425.9

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 17

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 110.000

A. POINT	X = 50.00	X = 150.00
1	174.7	-17.1
2	-988.2	-50.7
3	-495.1	-78.9
4	-373.2	-98.3
5	-337.3	-107.3
6	-340.3	-105.6
7	-361.4	-93.9
8	-395.0	-73.6
9	-437.7	-46.4
10	-489.4	-13.8
11	-551.1	23.0
12	-625.1	63.5
13	-716.5	107.9
14	-832.8	157.6
15	-987.1	215.4
16	-1203.7	287.4
17	-1530.1	385.7
18	-2081.2	540.1
19	-3057.7	802.7
20	-11543.2	3021.2

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 18

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 120.000

A. POINT	X = 50.00	X = 150.00
1	-2290.0	-32.1
2	43.0	-92.5
3	-127.4	-136.4
4	-175.5	-157.5
5	-213.8	-155.0
6	-249.6	-132.0
7	-287.9	-94.1
8	-329.9	-46.7
9	-377.6	5.3
10	-431.9	58.8
11	-495.9	112.1
12	-571.9	164.9
13	-665.4	218.1
14	-784.1	274.1
15	-941.6	337.2
16	-1161.6	415.1
17	-1492.4	522.5
18	-2048.9	695.0
19	-3032.2	993.3
20	-11536.8	3592.4

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 19

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 130.000

A. POINT	X = 50.00	X = 150.00
1	-2320.2	-73.5
2	9.0	-195.1
3	-125.7	-249.6
4	-176.2	-231.9
5	-212.8	-162.8
6	-248.4	-69.0
7	-284.4	28.6
8	-323.6	118.4
9	-368.9	195.9
10	-421.1	260.9
11	-482.6	315.7
12	-557.2	363.6
13	-649.9	408.4
14	-768.3	455.0
15	-926.2	509.3
16	-1148.2	581.1
17	-1481.7	687.6
18	-2043.8	869.5
19	-3036.4	1196.3
20	-11605.0	4165.9

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 20

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 140.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	144.1	-254.6
2	-813.6	-464.3
3	-565.0	-253.7
4	-429.6	68.9
5	-376.6	320.9
6	-360.1	478.3
7	-366.0	565.5
8	-386.9	608.8
9	-417.6	627.0
10	-460.8	632.3
11	-514.6	632.6
12	-583.8	633.3
13	-672.3	638.7
14	-788.3	653.4
15	-945.4	683.1
16	-1168.1	737.2
17	-1505.3	834.1
18	-2075.6	1017.4
19	-3084.6	1363.4
20	-11801.2	4631.3

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 21

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 148.500

A. POINT	X = 50.00	X = 150.00
1	-1330.4	588.3
2	-72.9	2995.8
3	-333.3	2300.9
4	-323.9	1804.6
5	-311.9	1486.4
6	-319.4	1266.2
7	-335.7	1105.2
8	-360.7	983.8
9	-396.3	890.8
10	-440.9	819.7
11	-496.7	766.9
12	-566.2	729.9
13	-655.5	708.5
14	-771.8	703.8
15	-929.0	719.5
16	-1151.8	763.6
17	-1489.1	853.4
18	-2058.8	1032.4
19	-3066.2	1376.4
20	-11761.0	4661.2

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 22

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 150.000

A. POINT	X = 50.00	X = 150.00
1	-1141.1	613.1
2	-212.6	2612.6
3	-378.5	2044.7
4	-345.9	1610.8
5	-324.2	1327.1
6	-325.8	1127.2
7	-339.3	978.4
8	-362.4	864.2
9	-396.6	775.0
10	-440.3	705.1
11	-495.3	651.3
12	-564.2	611.3
13	-653.1	584.3
14	-768.8	571.0
15	-925.7	573.5
16	-1148.0	597.4
17	-1484.9	654.8
18	-2054.2	776.5
19	-3060.9	1016.6
20	-11749.5	3364.9

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 23

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 160.000

A. POINT	X = 50.00	X = 150.00
1	-283.0	725.7
2	-845.6	875.9
3	-583.8	883.3
4	-445.7	732.2
5	-380.2	604.9
6	-354.5	496.8
7	-355.8	403.6
8	-370.2	322.0
9	-398.1	250.2
10	-437.5	185.8
11	-489.0	127.8
12	-555.1	73.7
13	-641.9	21.6
14	-755.3	-31.0
15	-910.4	-88.1
16	-1131.0	-155.6
17	-1466.3	-245.2
18	-2033.3	-382.9
19	-3037.3	-614.4
20	-11697.2	-2510.1

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 24

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 170.000

A. POINT	X = 50.00	X = 150.00
1	258.0	796.7
2	-1244.8	-219.0
3	-713.2	151.1
4	-508.6	178.4
5	-415.5	149.6
6	-372.6	99.4
7	-366.2	41.2
8	-375.2	-19.7
9	-399.1	-80.6
10	-435.8	-141.5
11	-485.1	-202.3
12	-549.4	-265.3
13	-634.9	-333.1
14	-746.8	-410.6
15	-900.8	-505.2
16	-1120.2	-630.4
17	-1454.6	-812.6
18	-2020.2	-1113.8
19	-3022.3	-1642.6
20	-11664.3	-6214.0

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 25

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 180.000

A. POINT	X = 50.00	X = 150.00
1	613.0	843.2
2	-1506.6	-937.5
3	-798.0	-329.4
4	-549.8	-185.0
5	-438.6	-149.2
6	-384.4	-161.3
7	-373.1	-196.5
8	-378.4	-244.0
9	-399.7	-297.7
10	-434.6	-356.3
11	-482.5	-418.9
12	-545.6	-487.7
13	-630.3	-565.9
14	-741.2	-659.6
15	-894.6	-778.8
16	-1113.2	-941.9
17	-1446.9	-1184.9
18	-2011.6	-1593.4
19	-3012.6	-2317.3
20	-11642.7	-8644.3

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 26

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 190.000

A. POINT	X = 50.00	X = 150.00
1	843.6	873.5
2	-1676.7	-1404.2
3	-853.2	-641.5
4	-576.6	-421.1
5	-453.7	-343.2
6	-392.1	-330.7
7	-377.5	-351.0
8	-380.5	-389.7
9	-400.1	-438.7
10	-433.9	-495.9
11	-480.8	-559.6
12	-543.2	-632.2
13	-627.3	-717.1
14	-737.6	-821.4
15	-890.5	-956.6
16	-1108.6	-1144.2
17	-1441.9	-1426.8
18	-2006.0	-1905.0
19	-3006.2	-2755.6
20	-11628.7	-10223.0

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.

PAGE 27

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 1 row with 20 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 200.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	931.5	885.0
2	-1741.6	-1582.0
3	-874.2	-760.4
4	-586.8	-511.1
5	-459.4	-417.2
6	-395.1	-395.2
7	-379.2	-409.9
8	-381.3	-445.2
9	-400.3	-492.4
10	-433.6	-549.1
11	-480.1	-613.2
12	-542.3	-687.3
13	-626.2	-774.8
14	-736.2	-883.0
15	-888.9	-1024.4
16	-1106.8	-1221.3
17	-1440.0	-1518.9
18	-2003.9	-2023.7
19	-3003.8	-2922.6
20	-11623.3	-10824.7

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 1.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Variable attachment. 1 row with 10 attach points per stringer.

PAGE 28

```

*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****

```

SKIN DATA:

```

=====

```

Young's modulus.....:	73800.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	3800.00
Lateral stiffener spacing.....:	500.00

APPLIED STRESSES:

```

=====

```

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

```

=====

```

	STIFFENER 1	STIFFENER 2
	-----	-----
X-coordinate.....:	50.00	150.00
No uniform attachment system:		
Number of points.....:	10	10
Thickness at attachment.....:	3.00	3.00
Mid line length.....:	50.00	50.00
Area.....:	150.00	150.00
Moment of inertia.....:	32250.00	32250.00
Distance from CG to outer cap:	25.00	25.00
Distance from CG to inner cap:	25.00	25.00

Output file: (cont.)

Program INTEGRAL. Example 1.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Variable attachment. 1 row with 10 attach points per stringer.

PAGE 29

```

*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****

```

ATTACHMENT POINTS FOR STIFFENER AT X = 50.00:

=====

NUMBER	X	Y	WIDTH
----	-----	-----	-----
1	50.00	1.50	3.00
2	50.00	4.50	3.00
3	50.00	7.50	3.00
4	50.00	10.50	3.00
5	50.00	13.50	3.00
6	50.00	16.50	3.00
7	50.00	19.50	3.00
8	50.00	22.50	3.00
9	50.00	25.50	3.00
10	50.00	28.50	3.00

ATTACHMENT POINTS FOR STIFFENER AT X = 150.00:

=====

NUMBER	X	Y	WIDTH
----	-----	-----	-----
1	150.00	1.50	3.00
2	150.00	4.50	3.00
3	150.00	7.50	3.00
4	150.00	10.50	3.00
5	150.00	13.50	3.00
6	150.00	16.50	3.00
7	150.00	19.50	3.00
8	150.00	22.50	3.00
9	150.00	25.50	3.00
10	150.00	28.50	3.00

Output file: (cont.)

Program INTEGRAL. Example 1.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Variable attachment. 1 row with 10 attach points per stringer.

PAGE 30

```
*****
*
*   <<< R E S U L T S   F O R   S K I N >>>
*
*****
```

APPLIED STRESSES:

```
-----
Average axial stress (YD).....:    200.00
Ratio of stresses (XD/YD).....:      0.50
Unit axial skin stress.....:    214.14
Unit lateral skin stress.....:    100.00
```

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	RESIDUAL STRENGTH	STRESS INTENSITY	STIFFENERS STATUS
-----	-----	-----	-----	-----	-----
0.00	214.14	1.00000	*****. **	0.00	II
10.00	214.14	0.99905	633.79	1199.13	II
20.00	214.14	0.99483	450.06	1688.66	II
30.00	214.14	0.98061	372.80	2038.61	II
40.00	214.14	0.93089	340.10	2234.64	II
48.50	214.14	0.69774	412.07	1844.35	II
50.00	214.14	0.76137	371.93	2043.42	TI
60.00	214.14	1.07781	239.84	3168.82	TI
70.00	214.14	1.29691	184.53	4118.48	TI
80.00	214.14	1.42642	156.94	4842.52	TI
90.00	214.14	1.48181	142.44	5335.70	TI
100.00	214.14	1.46528	136.65	5561.60	XI
110.00	214.14	1.40664	135.72	5599.61	XI
120.00	214.14	1.35411	134.99	5630.20	XI
130.00	214.14	1.29292	135.83	5595.28	XI
140.00	214.14	1.18196	143.17	5308.20	XI
148.50	214.14	0.83312	197.23	3853.43	XI
150.00	214.14	0.88978	183.74	4136.27	XT
160.00	214.14	1.17164	135.11	5625.12	XT
170.00	214.14	1.36678	112.36	6763.97	XT
180.00	214.14	1.48214	100.70	7547.49	XT
190.00	214.14	1.53147	94.85	8012.40	XT
200.00	214.14	1.51675	93.35	8141.54	XX

Beta factors refered to the unit axial skin stress.

Residual strength is based on the reference stress.

Key for stiffeners status: I=Intact
T=Transition
X=Failed

Example 2

In this example, the previous problem will be solved with the following additional considerations, in the two steps of the program:

– First step:

Forty attachment points will be considered for the transfer of the load between each stiffener and the skin, all of them aligned in two lines all along the stiffener and spaced uniformly at every 3 mm., with a distance between the lines of 1.5 mm. Consequently, in the data input, the uniform joint will be specified.

The load of each attachment point will be distributed in half of the thickness of the stiffener (1.5 mm.).

The results requested will be all the possible ones, that is, results for the skin, for the stiffeners and for the loads at all the attachment points.

– Second step:

Twenty attachment points will be considered for the transfer of the load between each stiffener and the skin.

In the data input, a variable joint will be specified, but with the same distribution as for the first step.

Results will only be requested for the skin.

The difference between both steps lies in the definition of the junction between stiffeners and the skin, although it will formally be the same in both cases, except for the number of attachment points considered.

Therefore, this example could be useful to evaluate the influence of the total number of attachment points per stringer.

With this data, we can prepare the input file. This file as well as the output file obtained by running the program are shown on next pages:

Input file:

```

1 Program INTEGRAL. Example 2.
  Crack in an integrally stiffened panel.
  Material: 2024 T351.
  Uniform attachment. 2 rows with 40 attach points per stringer.
2 SKIN DATA: E  μ  t  Kc  L
73800.0      0.33      2.0      3800.0      500.0
3 STRINGERS DATA: nl  X  ts  h  A  I  Zo  Zi
2      50.0      150.0
      3.0      3.0
      50.0      50.0
      150.0      150.0
      32250.0      32250.0
      25.0      25.0
      25.0      25.0
4 UNIFORM ATTACHMENT SYSTEM: nx  ny  Pitch  W
2 20 3.0 1.5
2 20 3.0 1.5
5 FAR FIELD STRESS: Reference_stress  Sx/Sy
200.00      0.50
6 CRACK LENGTHS
21 0.0  10.0  20.0  30.0  40.0  50.0
      60.0  70.0  80.0  90.0 100.0
      110.0 120.0 130.0 140.0 150.0
      160.0 170.0 180.0 190.0 200.0
7 PRINT CONTROL: Skin  Stringers  Forces
1  1  1
8 RUN THE PROGRAM

```

Input file: (cont.)

```

1 Program INTEGRAL. Example 2.
  Crack in an integrally stiffened panel.
  Material: 2024 T351.
  Variable attachment. 2 rows with 20 attach points per stringer.
4 VARIABLE ATTACHMENT: nu  X(i)  Y(i)  W(i)
  20      49.25      1.5      1.5
           50.75      1.5      1.5
           49.25      4.5      1.5
           50.75      4.5      1.5
           49.25      7.5      1.5
           50.75      7.5      1.5
           49.25     10.5      1.5
           50.75     10.5      1.5
           49.25     13.5      1.5
           50.75     13.5      1.5
           49.25     16.5      1.5
           50.75     16.5      1.5
           49.25     19.5      1.5
           50.75     19.5      1.5
           49.25     22.5      1.5
           50.75     22.5      1.5
           49.25     25.5      1.5
           50.75     25.5      1.5
           49.25     28.5      1.5
           50.75     28.5      1.5
  20     149.25      1.5      1.5
           150.75      1.5      1.5
           149.25      4.5      1.5
           150.75      4.5      1.5
           149.25      7.5      1.5
           150.75      7.5      1.5
           149.25     10.5      1.5
           150.75     10.5      1.5
           149.25     13.5      1.5
           150.75     13.5      1.5
           149.25     16.5      1.5
           150.75     16.5      1.5
           149.25     19.5      1.5
           150.75     19.5      1.5
           149.25     22.5      1.5
           150.75     22.5      1.5
           149.25     25.5      1.5
           150.75     25.5      1.5
           149.25     28.5      1.5
           150.75     28.5      1.5
7 PRINT CONTROL: Skin  Stringers  Allowables  Forces
  1  0  0
8 RUN THE PROGRAM

```

Output file:

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 2 rows with 40 attach points per stringer.

PAGE 1

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

SKIN DATA:

```
=====
```

Young's modulus.....:	73800.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	3800.00
Lateral stiffener spacing.....:	500.00

APPLIED STRESSES:

```
=====
```

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

```
=====
```

	STIFFENER 1	STIFFENER 2
	-----	-----
X-coordinate.....:	50.00	150.00
Uniform attachment system:		
Number of rows.....:	2	2
Number of points per row..:	20	20
Attachment pitch.....:	3.00	3.00
Attachment width.....:	1.50	1.50
Thickness at attachment.....:	3.00	3.00
Mid line length.....:	50.00	50.00
Area.....:	150.00	150.00
Moment of inertia.....:	32250.00	32250.00
Distance from CG to outer cap:	25.00	25.00
Distance from CG to inner cap:	25.00	25.00

Output file: (cont.)

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 2 rows with 40 attach points per stringer.

PAGE 2

```
*****
*
*   <<< R E S U L T S   F O R   S K I N >>>
*
*****
```

APPLIED STRESSES:

```
-----
Average axial stress (YD).....:    200.00
Ratio of stresses (XD/YD).....:      0.50
Unit axial skin stress.....:    214.14
Unit lateral skin stress.....:    100.00
```

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	RESIDUAL STRENGTH	STRESS INTENSITY	STIFFENERS STATUS
-----	-----	-----	-----	-----	-----
0.00	214.14	1.00000	*****. **	0.00	II
10.00	214.14	0.99741	634.83	1197.16	II
20.00	214.14	0.98825	453.06	1677.49	II
30.00	214.14	0.96692	378.08	2010.16	II
40.00	214.14	0.91168	347.27	2188.52	II
48.50	214.14	0.58153	494.41	1537.17	II
50.00	214.14	0.65224	434.15	1750.55	TI
60.00	214.14	1.00395	257.48	2951.67	TI
70.00	214.14	1.24746	191.85	3961.45	TI
80.00	214.14	1.39141	160.89	4723.65	TI
90.00	214.14	1.45296	145.26	5231.84	TI
100.00	214.14	1.43460	139.57	5445.13	XI
110.00	214.14	1.37930	138.41	5490.77	XI
120.00	214.14	1.32166	138.30	5495.26	XI
130.00	214.14	1.25322	140.13	5423.46	XI
140.00	214.14	1.15028	147.12	5165.91	XI
148.50	214.14	0.67173	244.61	3106.97	XI
150.00	214.14	0.74106	220.62	3444.91	XT
160.00	214.14	1.08588	145.78	5213.39	XT
170.00	214.14	1.32462	115.94	6555.30	XT
180.00	214.14	1.46574	101.82	7464.01	XT
190.00	214.14	1.52609	95.19	7984.28	XT
200.00	214.14	1.50808	93.88	8095.05	XX

Beta factors refered to the unit axial skin stress.

Residual strength is based on the reference stress.

Key for stiffeners status: I=Intact
T=Transition
X=Failed

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 3

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X =  50.00 >>>
*
*****
```

Unit stiffener stress: 181.14

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	STRESSES		
			OUTER CAP	AXIAL	INNER CAP
0.00	27171.4	0.0	181.14	181.14	181.14
10.00	27312.6	2390.2	183.94	182.08	180.23
20.00	27778.7	10488.5	193.32	185.19	177.06
30.00	28737.2	28125.4	213.38	191.58	169.78
40.00	30732.3	68681.5	258.12	204.88	151.64
48.50	36310.0	199309.5	396.57	242.07	87.56
50.00	0.0	0.0	0.00	0.00	0.00
60.00	0.0	0.0	0.00	0.00	0.00
70.00	0.0	0.0	0.00	0.00	0.00
80.00	0.0	0.0	0.00	0.00	0.00
90.00	0.0	0.0	0.00	0.00	0.00
100.00	0.0	0.0	0.00	0.00	0.00
110.00	0.0	0.0	0.00	0.00	0.00
120.00	0.0	0.0	0.00	0.00	0.00
130.00	0.0	0.0	0.00	0.00	0.00
140.00	0.0	0.0	0.00	0.00	0.00
148.50	0.0	0.0	0.00	0.00	0.00
150.00	0.0	0.0	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 4

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X = 150.00 >>>
*
*****
```

Unit stiffener stress: 181.14

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	STRESSES		
			OUTER CAP	AXIAL	INNER CAP
0.00	27171.4	0.0	181.14	181.14	181.14
10.00	27183.4	198.2	181.38	181.22	181.07
20.00	27219.3	795.2	182.08	181.46	180.85
30.00	27279.5	1796.1	183.26	181.86	180.47
40.00	27362.9	3188.6	184.89	182.42	179.95
48.50	27439.2	4472.7	186.39	182.93	179.46
50.00	27726.4	9190.1	191.97	184.84	177.72
60.00	29028.4	30570.2	217.22	193.52	169.83
70.00	29849.3	44049.1	233.14	199.00	164.85
80.00	30387.8	52893.3	243.59	202.59	161.58
90.00	30737.7	58638.7	250.37	204.92	159.46
100.00	30871.1	60828.1	252.96	205.81	158.65
110.00	31989.1	79554.6	274.93	213.26	151.59
120.00	33707.0	109305.1	309.45	224.71	139.98
130.00	36387.8	158671.3	365.59	242.59	119.58
140.00	40829.4	250098.1	466.07	272.20	78.32
148.50	54261.9	564717.1	799.51	361.75	-76.02
150.00	0.0	0.0	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 5

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 0.000

A. POINT	X = 50.00	X = 150.00
1	0.0	0.0
2	0.0	0.0
3	0.0	0.0
4	0.0	0.0
5	0.0	0.0
6	0.0	0.0
7	0.0	0.0
8	0.0	0.0
9	0.0	0.0
10	0.0	0.0
11	0.0	0.0
12	0.0	0.0
13	0.0	0.0
14	0.0	0.0
15	0.0	0.0
16	0.0	0.0
17	0.0	0.0
18	0.0	0.0
19	0.0	0.0
20	0.0	0.0
21	0.0	0.0
22	0.0	0.0
23	0.0	0.0
24	0.0	0.0
25	0.0	0.0
26	0.0	0.0
27	0.0	0.0
28	0.0	0.0
29	0.0	0.0
30	0.0	0.0
31	0.0	0.0
32	0.0	0.0
33	0.0	0.0
34	0.0	0.0
35	0.0	0.0
36	0.0	0.0
37	0.0	0.0
38	0.0	0.0
39	0.0	0.0
40	0.0	0.0

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 6

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 10.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	-0.2	0.0
2	-1.0	0.0
3	-0.5	0.0
4	-2.9	-0.1
5	-0.2	0.0
6	-4.7	-0.1
7	0.7	0.0
8	-6.4	-0.1
9	2.3	0.0
10	-7.8	-0.2
11	4.5	0.1
12	-8.9	-0.2
13	7.2	0.1
14	-9.8	-0.2
15	10.1	0.1
16	-10.5	-0.3
17	12.9	0.2
18	-11.1	-0.3
19	15.5	0.2
20	-11.4	-0.3
21	17.8	0.2
22	-11.6	-0.3
23	19.7	0.3
24	-11.6	-0.3
25	21.1	0.4
26	-11.6	-0.4
27	22.2	0.5
28	-11.4	-0.4
29	22.9	0.6
30	-11.1	-0.4
31	23.4	0.7
32	-10.7	-0.3
33	23.8	0.9
34	-10.0	-0.2
35	24.4	1.2
36	-8.9	-0.1
37	26.0	1.7
38	-7.1	0.2
39	46.0	5.2
40	10.0	3.4

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 7

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 20.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	-1.2	0.0
2	-4.9	-0.1
3	-2.4	0.0
4	-14.4	-0.2
5	-0.4	0.1
6	-23.1	-0.4
7	5.5	0.1
8	-30.5	-0.5
9	15.2	0.2
10	-36.5	-0.6
11	27.5	0.3
12	-41.1	-0.8
13	41.0	0.3
14	-44.5	-0.9
15	54.1	0.5
16	-46.9	-1.0
17	66.0	0.6
18	-48.4	-1.2
19	75.9	0.8
20	-49.3	-1.3
21	83.7	1.0
22	-49.6	-1.4
23	89.3	1.3
24	-49.4	-1.4
25	93.1	1.6
26	-48.7	-1.5
27	95.2	2.0
28	-47.7	-1.5
29	96.1	2.4
30	-46.2	-1.4
31	96.2	3.0
32	-44.1	-1.3
33	96.3	3.8
34	-41.3	-1.0
35	97.5	4.9
36	-37.0	-0.4
37	102.3	6.7
38	-30.0	0.9
39	176.3	20.7
40	33.3	13.5

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 8

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 30.000

A. POINT	X = 50.00	X = 150.00
1	-4.9	0.0
2	-17.1	-0.2
3	-7.3	0.1
4	-49.3	-0.5
5	7.8	0.2
6	-75.6	-0.8
7	40.0	0.3
8	-95.0	-1.1
9	81.5	0.4
10	-108.1	-1.5
11	123.9	0.6
12	-116.5	-1.8
13	161.2	0.8
14	-121.5	-2.1
15	190.6	1.1
16	-124.1	-2.4
17	211.8	1.5
18	-124.9	-2.7
19	225.4	1.9
20	-124.5	-2.9
21	232.7	2.4
22	-123.0	-3.1
23	235.2	3.0
24	-120.7	-3.3
25	234.0	3.7
26	-117.6	-3.4
27	230.3	4.6
28	-113.7	-3.4
29	225.1	5.6
30	-109.1	-3.3
31	219.5	6.9
32	-103.6	-3.0
33	214.7	8.6
34	-96.7	-2.3
35	212.9	11.1
36	-87.2	-1.0
37	219.1	15.3
38	-72.6	1.9
39	362.5	46.6
40	50.9	30.1

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 9

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 40.000

A. POINT	X = 50.00	X = 150.00
1	-26.1	0.1
2	-75.0	-0.3
3	18.8	0.2
4	-191.1	-0.9
5	179.1	0.4
6	-248.2	-1.5
7	339.7	0.6
8	-269.1	-2.1
9	447.5	0.9
10	-275.0	-2.7
11	505.2	1.2
12	-274.6	-3.2
13	528.3	1.6
14	-270.9	-3.8
15	530.0	2.2
16	-265.2	-4.3
17	519.2	2.8
18	-258.0	-4.8
19	501.1	3.6
20	-249.8	-5.2
21	479.4	4.6
22	-240.8	-5.7
23	456.1	5.7
24	-231.3	-6.0
25	432.6	6.9
26	-221.4	-6.2
27	409.6	8.5
28	-211.0	-6.2
29	388.2	10.3
30	-200.1	-6.0
31	368.9	12.6
32	-188.4	-5.5
33	353.0	15.6
34	-175.2	-4.3
35	343.4	19.9
36	-159.1	-1.9
37	346.6	27.2
38	-136.4	3.0
39	546.1	81.9
40	34.8	52.2

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 10

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 48.500

A. POINT	X = 50.00	X = 150.00
1	1424.9	0.1
2	-743.4	-0.4
3	1870.2	0.4
4	-684.9	-1.2
5	1559.9	0.7
6	-598.9	-2.1
7	1335.8	1.0
8	-549.2	-3.0
9	1168.8	1.5
10	-510.7	-3.8
11	1038.8	2.0
12	-477.9	-4.6
13	933.8	2.6
14	-448.7	-5.4
15	846.7	3.5
16	-422.2	-6.1
17	772.8	4.4
18	-397.8	-6.8
19	709.0	5.5
20	-375.1	-7.5
21	653.3	6.9
22	-353.8	-8.0
23	604.1	8.4
24	-333.7	-8.5
25	560.6	10.2
26	-314.6	-8.8
27	522.1	12.3
28	-296.3	-8.8
29	488.1	14.9
30	-278.5	-8.6
31	458.8	18.0
32	-260.6	-7.9
33	435.0	22.1
34	-241.9	-6.3
35	419.3	28.1
36	-220.7	-3.0
37	418.9	38.1
38	-193.7	3.7
39	633.1	113.1
40	-12.6	71.1

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 11

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 50.000

A. POINT	X = 50.00	X = 150.00
1	1357.6	0.2
2	-815.1	-1.3
3	1771.5	0.6
4	-716.3	-3.9
5	1492.6	1.2
6	-631.2	-6.5
7	1287.1	2.1
8	-587.2	-9.1
9	1133.5	3.5
10	-553.8	-11.5
11	1013.4	5.3
12	-525.8	-13.9
13	915.9	7.6
14	-501.2	-16.1
15	834.3	10.3
16	-479.1	-18.2
17	764.5	13.6
18	-458.9	-20.0
19	703.5	17.2
20	-440.3	-21.7
21	649.5	21.2
22	-423.2	-23.1
23	600.9	25.6
24	-407.4	-24.2
25	556.8	30.4
26	-392.9	-25.0
27	516.1	35.7
28	-379.9	-25.3
29	478.0	41.6
30	-368.7	-24.9
31	441.8	48.5
32	-359.8	-23.4
33	406.4	57.2
34	-354.5	-20.3
35	369.9	69.4
36	-355.6	-13.8
37	329.5	89.6
38	-370.8	-0.4
39	171.6	242.0
40	-573.5	134.5

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 2 rows with 40 attach points per stringer.

PAGE 12

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 60.000

A. POINT	X = 50.00	X = 150.00
1	1052.7	0.5
2	-1139.9	-5.2
3	1324.1	1.6
4	-858.7	-15.9
5	1187.7	3.5
6	-778.0	-26.5
7	1066.5	7.1
8	-759.3	-36.8
9	973.6	12.5
10	-749.3	-46.7
11	898.3	20.1
12	-742.8	-56.0
13	834.5	29.8
14	-738.9	-64.6
15	778.3	41.5
16	-736.7	-72.6
17	727.0	55.0
18	-735.9	-79.7
19	678.9	70.0
20	-736.1	-86.1
21	632.6	86.3
22	-737.8	-91.4
23	586.5	103.7
24	-741.3	-95.7
25	539.3	122.1
26	-747.8	-98.6
27	488.8	141.7
28	-759.0	-99.7
29	432.3	162.9
30	-777.8	-98.6
31	364.6	187.0
32	-809.6	-93.9
33	276.5	216.2
34	-864.6	-83.6
35	146.2	256.8
36	-966.8	-62.5
37	-75.3	323.2
38	-1173.6	-18.9
39	-1920.0	826.3
40	-3115.4	422.1

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 13

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 70.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	860.4	0.7
2	-1344.7	-7.7
3	1042.1	2.2
4	-948.4	-23.5
5	995.4	5.0
6	-870.4	-39.2
7	927.4	10.2
8	-867.8	-54.3
9	872.8	18.2
10	-872.5	-68.8
11	825.8	29.4
12	-879.6	-82.5
13	783.3	43.8
14	-888.7	-95.2
15	742.9	61.2
16	-899.1	-106.9
17	703.4	81.2
18	-910.5	-117.4
19	663.4	103.4
20	-922.6	-126.7
21	621.9	127.4
22	-936.1	-134.5
23	577.4	152.9
24	-951.8	-140.7
25	528.2	179.9
26	-971.5	-144.9
27	471.7	208.5
28	-997.9	-146.7
29	403.4	239.4
30	-1035.8	-145.0
31	316.0	274.2
32	-1093.1	-138.4
33	194.6	316.5
34	-1186.2	-123.6
35	5.2	374.9
36	-1352.2	-93.2
37	-330.6	470.5
38	-1679.8	-30.6
39	-3238.7	1194.7
40	-4718.0	603.4

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 14

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 80.000

A. POINT	X = 50.00	X = 150.00
1	734.3	0.9
2	-1479.0	-9.4
3	857.0	2.6
4	-1007.3	-28.5
5	869.3	6.0
6	-931.1	-47.4
7	836.1	12.2
8	-939.0	-65.8
9	806.6	21.9
10	-953.4	-83.3
11	778.2	35.5
12	-969.4	-99.9
13	749.6	53.0
14	-987.0	-115.3
15	719.7	74.1
16	-1005.7	-129.4
17	687.9	98.3
18	-1025.1	-142.1
19	653.3	125.2
20	-1045.0	-153.3
21	614.9	154.3
22	-1066.3	-162.8
23	571.5	185.2
24	-1090.0	-170.2
25	521.0	217.8
26	-1118.3	-175.4
27	460.4	252.3
28	-1154.7	-177.5
29	384.5	289.5
30	-1205.0	-175.5
31	284.0	331.5
32	-1279.1	-167.6
33	140.8	382.3
34	-1397.2	-149.8
35	-87.3	452.4
36	-1605.0	-113.3
37	-498.1	567.1
38	-2011.9	-38.2
39	-4103.9	1436.4
40	-5769.5	722.3

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 15

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 90.000

A. POINT	X = 50.00	X = 150.00
1	652.3	1.0
2	-1566.3	-10.4
3	736.8	2.8
4	-1045.5	-31.8
5	787.3	6.6
6	-970.6	-52.8
7	776.9	13.5
8	-985.2	-73.2
9	763.6	24.4
10	-1005.9	-92.8
11	747.3	39.5
12	-1027.7	-111.2
13	727.8	59.0
14	-1050.9	-128.3
15	704.7	82.5
16	-1074.9	-144.0
17	677.8	109.5
18	-1099.5	-158.2
19	646.7	139.4
20	-1124.5	-170.6
21	610.4	171.8
22	-1150.8	-181.1
23	567.6	206.2
24	-1179.7	-189.4
25	516.3	242.5
26	-1213.6	-195.1
27	453.1	280.8
28	-1256.6	-197.5
29	372.2	322.1
30	-1314.9	-195.3
31	263.3	368.7
32	-1400.0	-186.5
33	105.9	425.0
34	-1534.3	-166.8
35	-147.4	502.8
36	-1769.2	-126.4
37	-606.9	629.9
38	-2227.6	-43.2
39	-4666.0	1593.4
40	-6452.5	799.6

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 16

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 100.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	621.1	1.0
2	-1599.6	-10.9
3	691.0	2.9
4	-1060.1	-33.0
5	756.1	6.9
6	-985.6	-54.9
7	754.3	14.0
8	-1002.9	-76.1
9	747.3	25.3
10	-1026.0	-96.4
11	735.6	41.0
12	-1049.9	-115.5
13	719.5	61.2
14	-1075.2	-133.3
15	698.9	85.7
16	-1101.3	-149.6
17	674.0	113.7
18	-1127.9	-164.3
19	644.1	144.8
20	-1154.8	-177.2
21	608.6	178.5
22	-1183.0	-188.1
23	566.1	214.2
24	-1213.9	-196.7
25	514.5	251.9
26	-1250.0	-202.7
27	450.3	291.6
28	-1295.4	-205.1
29	367.5	334.5
30	-1356.8	-202.8
31	255.4	382.9
32	-1446.1	-193.7
33	92.6	441.3
34	-1586.6	-173.3
35	-170.3	522.0
36	-1831.8	-131.4
37	-648.3	653.8
38	-2309.8	-45.1
39	-4880.2	1653.3
40	-6712.9	829.0

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 17

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 110.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	1231.8	-0.1
2	-2407.6	-17.1
3	465.8	0.8
4	-1116.5	-51.4
5	601.7	6.1
6	-945.0	-84.9
7	642.8	18.6
8	-906.0	-116.7
9	652.7	39.3
10	-905.3	-146.4
11	648.1	68.1
12	-918.5	-173.4
13	636.0	104.1
14	-938.9	-197.9
15	618.5	145.9
16	-962.1	-219.6
17	596.7	191.6
18	-988.4	-238.4
19	569.8	239.7
20	-1017.0	-254.5
21	537.6	288.8
22	-1048.5	-267.6
23	498.3	338.1
24	-1084.3	-277.4
25	449.9	387.4
26	-1126.8	-283.7
27	388.9	437.0
28	-1179.9	-285.4
29	309.0	488.4
30	-1249.9	-281.4
31	199.3	544.4
32	-1348.6	-268.9
33	38.2	611.4
34	-1499.3	-242.8
35	-224.6	703.9
36	-1756.0	-191.1
37	-705.7	857.7
38	-2247.3	-86.3
39	-4983.9	2064.5
40	-6696.5	966.9

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 18

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 120.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	-98.3	-3.0
2	-2261.1	-30.1
3	607.7	-5.0
4	-585.9	-89.7
5	609.1	6.5
6	-711.2	-146.0
7	612.2	36.0
8	-773.6	-197.0
9	604.2	83.7
10	-807.0	-242.0
11	592.5	145.9
12	-832.6	-280.6
13	577.8	217.3
14	-856.6	-313.5
15	560.0	292.7
16	-881.3	-341.0
17	538.9	367.7
18	-907.7	-363.6
19	513.8	439.4
20	-936.9	-381.7
21	483.6	506.3
22	-969.4	-395.4
23	446.4	568.0
24	-1007.0	-404.8
25	400.1	625.1
26	-1052.1	-409.3
27	341.3	679.0
28	-1108.7	-408.3
29	263.0	732.4
30	-1183.1	-400.1
31	154.6	789.5
32	-1287.2	-381.8
33	-6.1	858.7
34	-1444.6	-347.5
35	-270.3	957.9
36	-1709.6	-282.8
37	-756.5	1130.7
38	-2212.5	-155.6
39	-5095.6	2574.0
40	-6721.7	1103.5

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 19

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 130.000

A. POINT	X = 50.00	X = 150.00
1	-909.0	-13.2
2	-1106.5	-63.5
3	642.9	-17.9
4	-686.4	-184.5
5	595.4	30.5
6	-748.8	-288.8
7	562.1	132.7
8	-757.7	-371.9
9	547.0	268.3
10	-775.5	-435.6
11	535.4	412.7
12	-795.1	-483.8
13	522.8	548.6
14	-815.8	-520.2
15	507.3	666.9
16	-838.0	-547.7
17	488.8	765.1
18	-862.5	-567.9
19	466.0	844.2
20	-889.7	-582.1
21	437.8	907.4
22	-921.4	-590.8
23	402.7	958.0
24	-958.7	-594.1
25	358.6	999.7
26	-1004.1	-591.8
27	301.2	1036.4
28	-1061.8	-583.1
29	224.4	1072.5
30	-1138.0	-566.4
31	116.7	1113.9
32	-1244.9	-538.5
33	-43.8	1170.6
34	-1406.2	-492.5
35	-309.4	1263.8
36	-1676.6	-412.3
37	-800.4	1444.1
38	-2188.0	-261.8
39	-5193.4	3102.9
40	-6749.0	1186.6

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 20

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 140.000

A. POINT	X = 50.00	X = 150.00
1	1067.0	-63.3
2	-2860.0	-199.2
3	338.2	45.4
4	-979.7	-516.2
5	430.1	440.1
6	-785.2	-689.4
7	473.9	846.8
8	-734.9	-772.3
9	490.1	1136.5
10	-725.7	-815.4
11	490.0	1311.9
12	-732.9	-839.8
13	482.3	1407.4
14	-747.0	-853.3
15	469.0	1452.9
16	-767.2	-859.1
17	450.8	1468.3
18	-789.5	-858.9
19	428.3	1466.5
20	-816.5	-853.6
21	400.2	1455.3
22	-847.7	-843.8
23	365.8	1440.3
24	-884.8	-829.7
25	322.0	1425.3
26	-930.6	-811.0
27	265.3	1413.3
28	-988.6	-786.8
29	189.4	1408.1
30	-1065.1	-755.6
31	83.3	1415.0
32	-1172.6	-713.8
33	-75.7	1443.4
34	-1333.8	-654.3
35	-338.5	1515.2
36	-1603.7	-560.5
37	-825.1	1683.8
38	-2111.7	-394.9
39	-5177.4	3426.6
40	-6623.2	1126.5

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 2 rows with 40 attach points per stringer.

PAGE 21

```

*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****

```

Half crack length: 148.500

A. POINT	X = 50.00	X = 150.00
1	-1644.2	3547.9
2	269.6	-1831.2
3	259.1	4637.0
4	-502.2	-1675.3
5	399.4	3711.2
6	-736.8	-1503.6
7	445.0	3147.8
8	-755.6	-1423.4
9	462.0	2785.8
10	-758.6	-1362.9
11	464.6	2528.0
12	-765.4	-1311.5
13	458.5	2331.2
14	-775.9	-1265.5
15	447.8	2174.8
16	-791.7	-1223.1
17	431.0	2047.0
18	-810.8	-1182.9
19	410.0	1941.1
20	-834.5	-1144.1
21	384.5	1852.7
22	-863.8	-1105.8
23	350.8	1779.2
24	-899.0	-1067.4
25	308.1	1719.0
26	-943.4	-1027.7
27	252.1	1671.8
28	-1000.8	-985.2
29	176.7	1638.9
30	-1077.6	-937.7
31	70.1	1624.1
32	-1185.8	-881.3
33	-90.2	1635.9
34	-1349.1	-808.4
35	-356.2	1696.3
36	-1623.2	-701.5
37	-850.1	1860.9
38	-2139.8	-522.6
39	-5277.5	3669.9
40	-6728.5	1051.0

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 22

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 150.000

A. POINT	X = 50.00	X = 150.00
1	-1424.7	3495.8
2	250.7	-1842.2
3	212.8	4352.7
4	-557.7	-1748.1
5	364.0	3520.7
6	-747.2	-1560.9
7	417.4	3012.1
8	-751.7	-1475.5
9	437.7	2683.5
10	-749.1	-1415.8
11	442.4	2448.0
12	-752.6	-1367.8
13	438.0	2267.2
14	-761.7	-1326.4
15	427.8	2122.5
16	-776.1	-1289.0
17	411.8	2003.5
18	-794.4	-1254.0
19	391.5	1903.9
20	-817.5	-1220.6
21	366.2	1819.9
22	-846.2	-1187.7
23	332.9	1748.8
24	-881.1	-1154.9
25	290.5	1689.3
26	-925.2	-1121.1
27	234.9	1640.7
28	-982.4	-1085.6
29	159.7	1603.7
30	-1059.1	-1046.6
31	53.5	1580.9
32	-1167.3	-1001.8
33	-106.7	1578.8
34	-1330.6	-945.9
35	-372.6	1613.4
36	-1604.8	-867.6
37	-866.8	1727.1
38	-2121.4	-742.9
39	-5297.1	3054.2
40	-6709.2	336.4

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 23

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 160.000

A. POINT	X = 50.00	X = 150.00
1	-430.2	3259.5
2	165.3	-1892.3
3	3.1	3064.1
4	-809.7	-2078.3
5	203.7	2657.2
6	-794.3	-1820.9
7	292.5	2396.9
8	-734.1	-1711.6
9	327.6	2219.6
10	-706.1	-1655.4
11	341.4	2085.5
12	-694.8	-1622.8
13	345.2	1977.1
14	-697.1	-1602.1
15	337.2	1885.6
16	-705.1	-1587.7
17	324.6	1806.1
18	-720.3	-1576.6
19	307.6	1735.3
20	-740.8	-1567.2
21	283.5	1671.0
22	-766.5	-1558.7
23	251.6	1611.2
24	-799.9	-1551.1
25	211.2	1554.6
26	-842.8	-1544.7
27	156.8	1499.4
28	-899.4	-1540.5
29	82.8	1443.9
30	-975.1	-1540.4
31	-21.8	1385.4
32	-1083.6	-1547.8
33	-181.3	1319.7
34	-1246.8	-1569.4
35	-446.7	1237.4
36	-1521.4	-1620.5
37	-942.0	1120.8
38	-2038.0	-1741.0
39	-5385.7	263.4
40	-6621.8	-2902.5

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 24

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 170.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	196.8	3110.6
2	111.4	-1923.8
3	-129.2	2251.7
4	-968.5	-2286.4
5	102.6	2112.9
6	-824.0	-1984.8
7	213.8	2009.1
8	-723.1	-1860.5
9	258.2	1927.2
10	-679.0	-1806.5
11	277.7	1856.9
12	-658.4	-1783.5
13	286.6	1794.2
14	-656.4	-1775.9
15	280.1	1736.3
16	-660.4	-1776.1
17	269.7	1681.7
18	-673.5	-1779.9
19	254.6	1629.0
20	-692.5	-1785.7
21	231.3	1577.1
22	-716.3	-1792.6
23	200.4	1524.5
24	-748.7	-1800.8
25	161.1	1469.8
26	-790.8	-1811.7
27	107.6	1410.3
28	-847.1	-1827.3
29	34.2	1343.1
30	-922.2	-1851.7
31	-69.3	1262.1
32	-1030.8	-1892.0
33	-228.4	1156.3
34	-1193.9	-1962.4
35	-493.5	1000.3
36	-1468.8	-2095.1
37	-989.5	738.5
38	-1985.5	-2370.3
39	-5441.5	-1496.0
40	-6566.7	-4944.5

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.

PAGE 25

Crack in an integrally stiffened panel.

Material: 2024 T351.

Uniform attachment. 2 rows with 40 attach points per stringer.

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 180.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	608.3	3012.9
2	76.1	-1944.5
3	-215.9	1718.7
4	-1072.7	-2422.9
5	36.2	1755.7
6	-843.4	-2092.3
7	162.1	1754.6
8	-715.8	-1958.2
9	212.6	1735.3
10	-661.2	-1905.6
11	236.0	1707.0
12	-634.4	-1889.0
13	248.2	1674.2
14	-629.7	-1890.0
15	242.6	1638.3
16	-631.0	-1899.6
17	233.7	1600.0
18	-642.9	-1913.4
19	219.9	1559.3
20	-660.7	-1929.1
21	197.1	1515.5
22	-683.3	-1946.1
23	166.8	1467.6
24	-715.1	-1964.7
25	128.3	1414.1
26	-756.7	-1986.9
27	75.3	1351.9
28	-812.8	-2015.5
29	2.4	1277.0
30	-887.4	-2056.0
31	-100.4	1181.2
32	-996.2	-2117.9
33	-259.3	1049.1
34	-1159.3	-2220.3
35	-524.1	844.8
36	-1434.3	-2406.5
37	-1020.6	487.7
38	-1951.0	-2783.1
39	-5478.2	-2650.4
40	-6530.5	-6284.3

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Uniform attachment. 2 rows with 40 attach points per stringer.

PAGE 26

```
*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
```

Half crack length: 190.000

A. POINT	X = 50.00	X = 150.00
1	875.5	2949.4
2	53.1	-1958.0
3	-272.3	1372.4
4	-1140.4	-2511.6
5	-6.8	1523.7
6	-856.1	-2162.2
7	128.5	1589.3
8	-711.1	-2021.6
9	183.1	1610.6
10	-649.7	-1970.0
11	208.8	1609.6
12	-618.9	-1957.5
13	223.2	1596.2
14	-612.3	-1964.1
15	218.3	1574.7
16	-611.9	-1979.9
17	210.3	1547.0
18	-622.9	-2000.0
19	197.4	1513.9
20	-640.1	-2022.2
21	174.9	1475.5
22	-661.9	-2045.8
23	145.0	1430.6
24	-693.2	-2071.2
25	106.9	1377.9
26	-734.5	-2100.7
27	54.3	1313.9
28	-790.5	-2137.7
29	-18.3	1234.1
30	-864.9	-2188.7
31	-120.7	1128.7
32	-973.7	-2264.6
33	-279.3	979.5
34	-1136.7	-2387.8
35	-544.0	743.7
36	-1411.9	-2608.8
37	-1040.8	324.8
38	-1928.6	-3051.4
39	-5502.0	-3400.4
40	-6507.0	-7154.7

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.
 Crack in an integrally stiffened panel.
 Material: 2024 T351.
 Uniform attachment. 2 rows with 40 attach points per stringer.

PAGE 27

```

*****
*
*   <<< F O R C E S   O N   A T T A C H M E N T   P O I N T S >>>
*
*****
  
```

Half crack length: 200.000

A. POINT	X = 50.00	X = 150.00
-----	-----	-----
1	977.4	2925.2
2	44.4	-1963.1
3	-293.8	1240.4
4	-1166.2	-2545.4
5	-23.3	1435.3
6	-860.9	-2188.8
7	115.7	1526.3
8	-709.3	-2045.8
9	171.8	1563.1
10	-645.3	-1994.5
11	198.5	1572.4
12	-613.0	-1983.6
13	213.7	1566.5
14	-605.7	-1992.3
15	209.0	1550.4
16	-604.7	-2010.5
17	201.4	1526.8
18	-615.4	-2033.1
19	188.8	1496.7
20	-632.3	-2057.7
21	166.4	1460.2
22	-653.8	-2083.8
23	136.7	1416.5
24	-684.9	-2111.8
25	98.8	1364.1
26	-726.1	-2144.1
27	46.3	1299.5
28	-782.0	-2184.3
29	-26.2	1217.7
30	-856.3	-2239.2
31	-128.4	1108.7
32	-965.1	-2320.6
33	-287.0	952.9
34	-1128.2	-2451.7
35	-551.6	705.2
36	-1403.3	-2685.9
37	-1048.5	262.7
38	-1920.1	-3153.6
39	-5511.1	-3686.2
40	-6498.1	-7486.3

Positive attachment forces induce compression in the skin.

Output file: (cont.)

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Variable attachment. 2 rows with 20 attach points per stringer.

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```
*****
*
*   <<< I N P U T   D A T A >>>
*
*****
```

SKIN DATA:

```
=====
```

Young's modulus.....:	73800.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	3800.00
Lateral stiffener spacing.....:	500.00

APPLIED STRESSES:

```
=====
```

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

```
=====
```

	STIFFENER 1	STIFFENER 2
	-----	-----
X-coordinate.....:	50.00	150.00
No uniform attachment system:		
Number of points.....:	20	20
Thickness at attachment.....:	3.00	3.00
Mid line length.....:	50.00	50.00
Area.....:	150.00	150.00
Moment of inertia.....:	32250.00	32250.00
Distance from CG to outer cap:	25.00	25.00
Distance from CG to inner cap:	25.00	25.00

Output file: (cont.)

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Variable attachment. 2 rows with 20 attach points per stringer.

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```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

ATTACHMENT POINTS FOR STIFFENER AT X = 50.00:

=====

NUMBER	X	Y	WIDTH
----	----	----	----
1	49.25	1.50	1.50
2	50.75	1.50	1.50
3	49.25	4.50	1.50
4	50.75	4.50	1.50
5	49.25	7.50	1.50
6	50.75	7.50	1.50
7	49.25	10.50	1.50
8	50.75	10.50	1.50
9	49.25	13.50	1.50
10	50.75	13.50	1.50
11	49.25	16.50	1.50
12	50.75	16.50	1.50
13	49.25	19.50	1.50
14	50.75	19.50	1.50
15	49.25	22.50	1.50
16	50.75	22.50	1.50
17	49.25	25.50	1.50
18	50.75	25.50	1.50
19	49.25	28.50	1.50
20	50.75	28.50	1.50

Output file: (cont.)

Program INTEGRAL. Example 2.

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Crack in an integrally stiffened panel.

Material: 2024 T351.

Variable attachment. 2 rows with 20 attach points per stringer.

```
*****
*
*   <<< I N P U T   D A T A >>>
*
*****
```

ATTACHMENT POINTS FOR STIFFENER AT X = 150.00:

=====

NUMBER	X	Y	WIDTH
----	----	----	----
1	149.25	1.50	1.50
2	150.75	1.50	1.50
3	149.25	4.50	1.50
4	150.75	4.50	1.50
5	149.25	7.50	1.50
6	150.75	7.50	1.50
7	149.25	10.50	1.50
8	150.75	10.50	1.50
9	149.25	13.50	1.50
10	150.75	13.50	1.50
11	149.25	16.50	1.50
12	150.75	16.50	1.50
13	149.25	19.50	1.50
14	150.75	19.50	1.50
15	149.25	22.50	1.50
16	150.75	22.50	1.50
17	149.25	25.50	1.50
18	150.75	25.50	1.50
19	149.25	28.50	1.50
20	150.75	28.50	1.50

Output file: (cont.)

Program INTEGRAL. Example 2.
Crack in an integrally stiffened panel.
Material: 2024 T351.
Variable attachment. 2 rows with 20 attach points per stringer.

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```
*****
*
*   <<< R E S U L T S   F O R   S K I N >>>
*
*****
```

APPLIED STRESSES:

```
-----
Average axial stress (YD).....:    200.00
Ratio of stresses (XD/YD).....:      0.50
Unit axial skin stress.....:    214.14
Unit lateral skin stress.....:    100.00
```

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	RESIDUAL STRENGTH	STRESS INTENSITY	STIFFENERS STATUS
-----	-----	-----	-----	-----	-----
0.00	214.14	1.00000	*****. **	0.00	II
10.00	214.14	0.99901	633.82	1199.08	II
20.00	214.14	0.99455	450.19	1688.18	II
30.00	214.14	0.97923	373.33	2035.74	II
40.00	214.14	0.92361	342.78	2217.16	II
48.50	214.14	0.58135	494.57	1536.70	II
50.00	214.14	0.65475	432.49	1757.28	TI
60.00	214.14	1.01982	253.47	2998.33	TI
70.00	214.14	1.27258	188.06	4041.23	TI
80.00	214.14	1.42200	157.43	4827.50	TI
90.00	214.14	1.48589	142.05	5350.41	TI
100.00	214.14	1.46683	136.51	5567.47	XI
110.00	214.14	1.40770	135.62	5603.84	XI
120.00	214.14	1.35407	134.99	5630.01	XI
130.00	214.14	1.29053	136.08	5584.95	XI
140.00	214.14	1.17771	143.69	5289.10	XI
148.50	214.14	0.66874	245.71	3093.13	XI
150.00	214.14	0.73928	221.15	3436.62	XT
160.00	214.14	1.09012	145.21	5233.73	XT
170.00	214.14	1.33302	115.21	6596.89	XT
180.00	214.14	1.47661	101.07	7519.34	XT
190.00	214.14	1.53801	94.45	8046.65	XT
200.00	214.14	1.51969	93.17	8157.34	XX

Beta factors refered to the unit axial skin stress.

Residual strength is based on the reference stress.

Key for stiffeners status: I=Intact
T=Transition
X=Failed